Sahaj Bhadja

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Summary

Associate Software Engineer with 2+ years of experience building **education platforms**, **Java/Python services**, React/TypeScript interfaces, and **AI-assisted workflows** supported by cloud delivery and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **data validation and reconciliation**, error handling, cloud storage, CI/CD releases, and stable production monitoring for dependable internal workflows reliably.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant education platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Engineered an **AI-powered education platform** using GPT-4o, Claude, Qwen-VL, and OCR to evaluate 10K+ submissions with 90%+ scoring consistency across text and handwritten assessment workflows at production scale.
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **automated testing and deployment**, implementing JWT access controls, WebSocket updates, Azure monitoring, Docker/Jenkins pipelines, and ARM templates across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Streamlined project delivery by 22% through a **Spring Boot API and Firebase integration**, applying unit automation, version control, and cross-platform workflows while collaborating directly with teammates and clients.

Skills

Languages: Java, TypeScript, JavaScript, Python, SQL, C++, Bash
Full Stack: React, Node.js, Express.js, Spring Boot, FastAPI, Flask, REST APIs, WebSockets, Microservices
AI & Data: OpenAI API, Claude, GPT-4o, LLM Systems, Transformers, FAISS, PostgreSQL, MySQL, MongoDB
Cloud & Delivery: AWS, Azure, Docker, Kubernetes, Jenkins, CI/CD, Git, GitHub
Quality & Engineering: Unit Testing, Integration Testing, PyTest, Mockito, Debugging, Observability, Agile, Algorithms

Projects

Big-Brain
- Built a **semantic retrieval system** over 50K+ documents using transformer embeddings and FAISS, delivering sub-second contextual search through optimized indexing, metadata storage, and similarity pipelines for question answering.

Chat2SQL
- Built a **Java and TypeScript SQL workspace** supporting 10+ databases, implementing schema-aware query generation, validation, Dockerized workflows, and structured execution with 82% accuracy across relational database environments.

FinNexus | Live
- Built a **secure TypeScript fintech platform** supporting 50K+ users and 1M+ transactions through Plaid integration, token-based access controls, real-time synchronization, Kubernetes deployment, and MongoDB-backed filtering workflows reliably.